GEORGIA INSTITUTE OF TECHNOLOGY
SCHOOLS OF CS AND ECE
ECE 4100/6100, CS 4290/6290 (CONTE)
REVIEW FOR EXAM 2 TOPIC LIST

1. **[Everything from Exam 1]**
2. **Pipelining (N4)**
   a) Example Pipeline: what IF, ID, EX, MEM, WB do
   b) Pipeline speedup calculation
   c) CPI = 1 + stalls, where stalls is average stall cycles per instruction
   d) How Program Characteristics + Pipeline Hazards => Stalls
   e) Structural hazards: what they are, how to avoid them
   f) Data Hazards & Solutions
      i) Data dependencies (pure, anti-, output-dependencies) vs. Data hazards (RAW, WAR, WAW) – know difference between these!
      ii) Pipeline forwarding (also called "register file bypass")
      iii) When forwarding doesn't work: Load-use and delay slots
   g) Control Hazards & Solutions
      i) Issues in WHEN and WHERE
         a) Delay slots, Squash slots w/likely bits
         b) Branch target buffer (BTB) & Hardware return address stack
      ii) Issues in WHETHER
         a) Software: Heuristics (eg, backward taken, forward not taken)
         b) Hardware branch prediction
            (1) 1-bit (valid bit in BTB)
            (2) Smith 2-bit counter
            (3) Gselect, Gshare
            (4) Yeh/Patt
            (5) Perceptron
3. **Instruction-Level Parallelism (N5)**
   a) Limitations of pipelining – going beyond CPI = 1
   b) Architecture of superscalar processor: Fetch, Dispatch, Schedule, Execute(FUs), and State Update
   c) Performance: retired IPC
   d) Dispatch and scheduling
      i) FICO algorithm (in-order issue)
         a) FICO example execution
         b) How FICO issue enforces dependencies
         c) What causes FICO to be too conservative and thus miss parallelism opportunities
      ii) FOCO algorithm (out-of-order issue)
         a) Full Tomasulo algorithm
         b) FOCO example
         c) The behaviors and usage of Dispatch, Scheduling, FU's, CDB, and register file
         d) RAT/Preg approach: how it operates
   e) State update
      i) What the problem is: imprecise interrupts
      ii) Reorder buffer with bypass
      iii) Future file
      iv) Using the ROB to free Pregs in RAT/Preg
      v) Checkpoint repair
4. Homeworks 1-4; Projects 1 & 2